

Bubble Sort

unsorted array.

7	2	12	8	3
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0 1 2 3 4

$7 > 2$, swap

2	7	12	8	3
---	---	----	---	---

$7 > 12$, no, no swap

2	7	12	8	3
---	---	----	---	---

$12 > 8$, yes, swap

2	7	8	12	3
---	---	---	----	---

$12 > 3$, yes, swap

2	7	8	3	12
---	---	---	---	----

$2 > 7$, no

$7 > 8$, no

2	7	8	3	12
---	---	---	---	----

$8 > 3$, yes, swap

2	7	3	8	12
---	---	---	---	----

$7 > 3$, yes, swap

2	3	7	8	12
---	---	---	---	----

sorted array.

Pseudocode & Time Complexity

int bubble_sort(int arr[])

{

int n ← arr.length; — ①

int temp; — ①

for (int i = n-1; i >= 1; i--)

{
↳ 1 + n + n-1 = ②n

for (int j = 0; j < i; j++)

{
↳ n-1 + x + 1 + x ②x+n

if (arr[j] > arr[j+1])

↳ ③x

{

// swap

temp = arr[j]; → ②x

arr[j] = arr[j+1]; → 3x

arr[j+1] = temp; → 2x

}
}

return arr; — ①

3

$$1 + 1 + 2n + 2x + \cancel{n} + 3x + 2x + 3x + 2x + \cancel{1}$$

$$12x + 3n + 3$$

$$= 3n + 12x + 3 \quad \text{where } x = 1 + 2 + 3 + \dots + n-1$$

$$= \frac{k(k+1)}{2}$$

$$= 3n + 12n \frac{(n+1)}{2} + 3$$

$$= \frac{n-1+n}{2}$$

$$= 3n + \frac{12n^2}{2} + \frac{12n}{2} + 3$$

$$\boxed{\frac{(n-1)n}{2}}$$

$$= \boxed{O(n^2)} \text{ Time Complexity.}$$